Current Reflections

CNN-LSTM development and backtesting | 16 October 2026

1. Improving the code and validation process

I asked ChatGPT to review my code and have since fixed numerous bugs. Many were simple typos, but others had more serious implications for the validity of my results. These included data leakage in the validation process and the loss of the first 41 validation days because the model lacked sufficient historical input. I corrected these issues and reorganized the code into clearer modules.

This process helped me distinguish between routine implementation work and the research decisions that require my attention. Tasks such as organizing functions across files can be tedious and time-consuming, so I use ChatGPT to assist with them. This allows me to spend more time understanding the model, checking the methodology, and interpreting results. I still need to understand and verify the changes it suggests.

I also corrected the CNN kernel size. With a 42-day input window, a kernel size of 42, a stride of 1, and no padding would collapse the convolution output to a single time step, leaving the LSTM with no meaningful sequence to model. Using a kernel size of 3 instead lets each convolution filter capture local patterns across three consecutive trading days while preserving a sequence of features for the LSTM after pooling.

2. Investigating the long runtime

The full backtest is estimated to take around six hours, so I explored whether the cloud resources available through my ChatGPT Pro subscription could speed it up. However, the cloud environment we checked had no available GPU, so I have not moved the backtest there. My local run is still in progress. Any performance improvement should preserve the training setup and validation methodology, and its benefit should be measured rather than assumed.

3. Next steps

With the basic pipeline in place and the identified issues addressed, my next step is to complete the CNN-LSTM backtest and evaluate its performance on the validation periods. I will use these results as a baseline before exploring ensemble models. This should help me assess whether combining models improves performance under the same evaluation procedure.